Max. Marks: 0

LAB ASSIGNMENT # 01

CIFAR-10 Image Classification using ANN/MLP + Hyperparameter Tuning

Objective

To gain hands-on experience in implementing and tuning Artificial Neural Networks (ANN/MLP) for **image classification** using the **CIFAR-10 dataset** and hyperparameter tuning techniques such as Grid Search, Random Search, and Optional Bayesian Optimization.

You will train and evaluate these models on the CIFAR-10 dataset, analyze metrics, and visualize attention effects.

Dataset

Use the CIFAR-10 dataset (10 classes, 32×32 color images). You may use the TensorFlow/Keras built-in loader.

Ensure:

- Images are normalized (e.g., scaled to [0,1]).

- Labels are one-hot encoded.

- Train/test split uses standard CIFAR-10 split (or a validated custom split).

Tasks:

1. Data Preparation: Load and preprocess CIFAR-10 (normalization, one-hot labels, optional augmentation).
2. Model Implementation: ANN with three Hidden Layers.
3. Model Training: Train ANN model with hyperparameters; save histories.
4. Model Evaluation: Evaluate test accuracy/loss; confusion matrices.
5. Performance Comparison: Plot training/validation curves; compare parameters and training time.
6. Report: Document preprocessing, architectures, training, evaluation, and discussion.

Deliverables

- Source code (preprocessing, model definitions, training, evaluation).
- Training & validation loss/accuracy plots.
- Confusion matrices and a table summarizing test accuracy and parameter counts.
- A concise report or Summary including all required sections.
- Notebook (Jupyter/Colab) or .py script.

Additional Challenges (Optional)

- Compare activation functions (ReLU, Sigmoid, Tanh)
- Compare optimizers (Adam vs SGD vs RMSProp)
- Try CNN instead of ANN and compare accuracy
- Implement cross-validation (Optional)

- Experiment with feature augmentation